

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Fifth Semester B.E. (E&C) (Credit System) Degree Examinations
January 2023

20ECE46 - INTRODUCTION TO SENSORS AND ACTUATORS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) With a neat diagram, explain the working of a 4 – bit binary absolute encoder for measurement of angular position. | 10 | L*2 | 2 | 1 |
| b) Define gauge factor of a strain gauge. Derive the expression for gauge factor of a strain gauge by approximating the metal grid lines by a single rectangular block. | 10 | L2 | 2 | 1 |
| 2. a) Identify an inductive sensor that can be used to measure direction sensitive linear displacement with neat diagram, explain the construction and working principle of the same. | 10 | L3 | 2 | 1 |
| b) Explain the working principle of a thermocouple. | 4 | L2 | 2 | 1 |
| c) A standard two – junction thermocouple configuration is being used to measure the temperature in a wind tunnel. The reference junction is held at a constant temperature of 20°C. A thermocouple table referenced to 0°C is available. A portion of the table is shown. | | | | |

Junction Temperature (°C)	Output Voltage (mV)
0	0
10	0.507
20	1.019
30	1.536
40	2.058
50	2.585
60	3.115
70	3.649
80	4.816
90	4.725
100	5.268

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|--|----|----|---|---|
| Determine the output voltage when the measuring junction is exposed to an air temperature of 90°C. | 6 | L3 | 2 | 1 |
| 3. a) Explain the working of a MEMS sensor for measurement of acceleration. | 5 | L2 | 2 | 1 |
| b) Define the following parameters with respect to sensors.
i) Accuracy
ii) Repeatability
iii) Reproducibility
iv) Precision
v) Sensitivity | 10 | L2 | 1 | 1 |
| c) With respect to strain gauges, explain how the effect of resistance change due to temperature variation can be compensated. | 5 | L2 | 2 | 1 |

P.T.O.

Unit – II

- | | | | | | |
|-------|--|----|----|---|---|
| 4. a) | Derive the equation and draw the speed – torque characteristics of a permanent magnet DC motor. | 10 | L2 | 3 | 1 |
| b) | With neat diagram, explain the method of PWM based closed loop control of DC motor speed. | 10 | L2 | 3 | 1 |
| 5. a) | With neat diagram, explain the construction, working and application of
i) Hydraulic pressure regulator
ii) Pilot operated spool valve | 10 | L2 | 4 | 1 |
| b) | Explain the components of a pneumatic system with neat diagram. | 10 | L2 | 4 | 1 |
| 6. a) | With field coil schematic and full step phase sequence, explain the construction and working of standard unipolar stepper motor. | 10 | L2 | 3 | 1 |
| b) | State and explain any five salient questions a designer may need to consider when choosing a motor for an application. | 10 | L2 | 3 | 1 |

Unit – III

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|-------|--|----|----|---|---|
| 7. a) | Draw the circuit of a difference amplifier. | 5 | L2 | 5 | 1 |
| b) | Discuss the advantages of an instrumentation amplifier over difference amplifier. | 6 | L2 | 5 | 1 |
| c) | Draw the circuit of an instrumentation amplifier and derive the equation of the output voltage with respect to the input voltages. | 9 | L2 | 5 | 1 |
| 8. a) | Discuss the need of an opto-isolator in sensor applications. Explain the working of any one type. | 10 | L2 | 5 | 1 |
| b) | Draw the circuit diagram and derive the expression for output voltage for an inverting and a non-inverting amplifier. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
